Tier 1 debt overlay: modeling notes

Sub-sovereign / quasi-sovereign stocks for six countries, to accompany the sovereign interest-burden model (2026 vintage). Companion CSVs: canada_tier1_provincials, australia_tier1_semis, china_tier1_local, germany_tier1_laender, france_tier1_agencies, italy_tier1_quasi. Fiscal years: Canada Mar-31, Australia Jun-30, others calendar. Prepared 29 Oct 2026.

Country	Stock (2025)	Tier 1 / Federal	10y CAGR	WAM	Spread (tc/now)	Character
China	RMB 70.8tn	~200%	16%	7.9y	+57 / +29	Dominant; policy-administered
Australia	A$606bn	~63%	10.3%	6.6y	+51 / +60	Fastest organic grower
Canada	C$934bn	~60%	4.7%	11.0y	+70 / +47	Classic federal-provincial
Germany	€625bn	~60%	-0.3%	6.2y	+36 / +50	Shrank; debt-brake bound
France	€370bn	~14%	4.0%	4.7y	+24 / +21	Quasi-OAT; shortening
Italy	€365bn	~13%	1.3%	2.4y	+12 / +7	Putable postal stock

Spread = assumed flat add to the federal curve, bp, through-cycle / current (Oct 2026). WAM and spread are 2025-debt-weighted. Punchline: the federal-only model most understates the interest trajectory for China (~3x), Australia (~1.6x) and Canada (~1.5x); Germany barely moves the slope; Italy and France are about structure (putability, bill-funding), not level.

Canada — provinces (ON, QC, AB, BC)

C$615bn (FY2016) → C$934bn (FY2025), CAGR ~4.7%. WAM ~11.0y, spread ~70bp through-cycle (~45–50bp at current cyclical tights).

- **Deficits don't explain the debt — don't drive issuance off the balance.** Cumulative balances 2016–25 are only ~–C$105bn against ~C$320bn of debt growth: provincial capital spending sits outside the accrual budget balance, so borrowing requirement ≈ deficit + ~C$20–30bn/yr of net capital. Calibrate the overlay to the debt-stock CAGR, not reported balances, or Tier 1 growth is understated by ~2/3.
- **Long and expensive relative to the federal ladder.** Four provinces ≈ C$930bn vs ~C$1.5tn federal marketable — a ~60% add — at WAM ~11y (vs federal 5.7y) and +70bp. It reprices slowly (UK-like) and damps the bear-case front-end benefit: Canada's 49% bill share drops sharply once provinces are blended in (provincial programmes run ~10–15% bills).
- **Spread differentiation:** ON 70 / QC 72 / AB 78 / BC 65 through-cycle. BC taxpayer-supported debt is tripling FY2023→FY2029 and has converged up to Ontario; Quebec 30s recently traded 5–6bp wider than Ontario on political risk, widest in a decade. A dynamic spread lever of ±15bp covers the realistic non-crisis range; the COVID spike (~+60–100bp, compressed by the BoC's PBPP) is the tail case.
- **Measures differ by province** (ON OFA total debt incl. OEFC; QC gross debt excl. Hydro-Québec's ~C$50bn of separately issued guaranteed paper; AB taxpayer-supported; BC total). Hydro-Québec is an optional add-on.

Australia — semis (NSWTC, TCV, QTC, WATC, SAFA, TasCorp, NTTC, ACT)

A$250bn (FY2016) → A$606bn (FY2025), CAGR ~10.3% — the fastest-growing Tier 1 stock anywhere. WAM ~6.6y, spread ~51bp through-cycle, ~60bp current.

- **Proportionally huge and still growing.** ~63% of the AGS stock, adding ~21pp of GDP to Australia's 35% federal ratio; the FY26 aggregate funding task exceeds A$100bn gross. The Int/GDP line for Australia roughly 1.6x's once semis are folded in.

– **The deficit–debt disconnect is even more extreme than Canada.** Cumulative operating deficits 2016–25 ≈ A$62bn vs ~A$356bn of debt growth: states run infrastructure booms below the operating line and CFAs on-lend to state corporations. The reported balance explains <20% of issuance — calibrate to the debt CAGR, full stop.

– **WAM ≈ the AGS ladder** (6.6y vs 6.1y), unlike Canada — so the overlay inherits roughly federal repricing speed (WATC the short outlier at ~4.8y). Issuance is benchmark-bond-heavy with minimal bills, so semis feel the bear-case long end more than the AGS programme does.

– **Credit-driven dispersion has widened structurally:** pre-COVID decade medians ~33–37bp; 2024 saw Victoria 10s at 90bp (AA, worst metrics, A$86bn refi wall 2029–34) vs WATC at 49bp (AAA, surpluses). Ranking: WA 42 < NSW 48 < Vic 52 ≈ SA 52 < Qld 55 < NT 65. One lever: 50bp ±20bp band; March-2020 (~+45bp pre-RBA purchases) is the tail.

– **Balances are GG net operating balances;** for a cash borrowing requirement add state asset-investment programs (~A$60–70bn/yr aggregate) net of depreciation.

China — LGBs + LGFV (chengtou) bonds

RMB 16.8tn (2016) → RMB 70.8tn (2025), CAGR ~16%. Weighted WAM ~7.9y, spread ~57bp through-cycle, ~29bp at today's all-time tights. Balance column = net issuance (Δ stock).

- **The biggest Tier 1 add by far — the local stock is ~2x the central one.** CGBs ≈ 30% of GDP; LGBs alone are ~40% and chengtou bonds another ~11%. Combined bond stock goes 30% → ~80%+ of GDP and China's Int/GDP path roughly triples.
- **Model the two series jointly or the swaps double-count.** The 2015–18 LGB explosion was mostly the loan-to-bond swap; 2024–28 growth includes the RMB 10tn hidden-debt swap, which retires LGFV paper as LGBs are issued (LGFV net issuance turns negative 2024–26). Only the combined aggregate's growth (~12–13%/yr, decelerating) is real net borrowing. The swap is also a WAM-extension and cost-reduction machine: ~4.5% / 2.2y chengtou debt becomes ~2% / 15y+ LGBs — the opposite of everyone else's refinancing gap.
- **WAM is bifurcated, not blended.** LGBs: stock ~9.5y and lengthening (Q1-2025 issuance averaged 17.2y at 1.97% — terming out at generational lows into a captive buyer base; mirror image of US bill reliance). LGFVs: ~2.2y rollover machines. A single 8y / ~2% blend is defensible, but the two halves' refi dynamics are opposite.
- **Spreads are administered, not market-clearing.** LGB issuance was window-guided (CGB+40 in 2018, ~+10–15 now); chengtou spreads collapsed ~200bp after the 2023 swap announcement. The bear case is not a term-premium rebuild — it is the implicit guarantee being tested (2018, 2022 Zunyi: AA chengtou 300–450bp). A credit-spread lever captures this; the curve lever does not.
- **Bonds are ~30% of true LGFV liabilities** (total interest-bearing ~RMB 55–60tn incl. loans/trust), plus ~US$115bn offshore chengtou (net financing negative four years; NDRC capping yields). Bond-only is consistent with the marketable-securities methodology — just know the augmented picture is ~2x bigger again. LGFV outstanding depends on vehicle definition (Wind ~11–12tn vs iFinD ~16.5tn); the series here is broad, bucket split (≈35/30/35) approximate; LGB numbers are hard MoF data.

Germany — Länder (16 states)

€641bn (2016) → €625bn (2025e), CAGR –0.3% — the only Tier 1 stock in the sample that shrank. WAM ~6.2y, spread ~36bp through-cycle, ~50bp now.

- **The mirror image of everyone else.** Länder deleveraged 2016–19 under a debt brake that binds harder than the Bund's (zero structural deficit until the 2025 reform), spiked in COVID, repaid, and only turned in 2024. ~60% add to the Bund stock but with no growth engine: the 2025 reform grants 0.35% of GDP (~€16bn/yr) of structural room, and the €500bn infrastructure fund is federal debt. Forward Tier 1 growth ~2–3%/yr, not 5–10%.
- **A securities screen finds ~€450bn, not €607bn.** Only ~74% of Länder debt is securities; the rest is Schuldschein and bank loans. For interest expense use the full stock (loans price off the same curve + similar spread); for a securities-only pull, scale accordingly.
- **The spread is a Bund-scarcity artifact, not credit.** Länder trade roughly flat to swaps; the pickup vs Bunds went ~20–30bp → 45–60bp mostly because Bund ASW collapsed. Mechanically the cost over the federal curve genuinely widened, but it is a regime shift likely to persist while Bund net supply is heavy — not mean-reverting credit. Cross-Länder dispersion is tiny (±10bp; Bremen/Saarland widest) given the solidarity principle and existing federal restructuring aid.
- **Small deficit–debt disconnect:** Länder budgets are cash-comprehensive, so fiscal balances track debt change reasonably well — one less adjustment. Aggregate and city-state anchors are hard Destatis numbers; per-issuer histories are 2024 shares scaled along the aggregate path (±€3–5bn); municipalities (€174bn, mostly loans) excluded.

France — agencies (CADES, Unédic, ACOSS, SFIL/CAFFIL, Bpifrance, AFL, IDFM)

€260bn (2016) → €370bn (2025), CAGR ~4.0%. Weighted WAM ~4.7y, weighted spread ~24bp — the cheapest and shortest stock in the sample. Balance column = net debt change.

- **Split the list into two buckets.** CADES + Unédic + ACOSS (~€248bn) are genuine fiscal burden. SFIL/CAFFIL + Bpifrance (~€106bn) are development banks whose debt is serviced by matched loan books — including them in an interest-expense overlay double-counts. Base case: CADES+Unédic+ACOSS (~8.5pp of GDP on top of the 94% central stock); banks as a sensitivity.
- **The flat aggregate hides a dangerous rotation.** CADES amortizes €16bn/yr (legislated, CRDS/CSG-funded, sunset 2033) while ACOSS piles up bills: the social-security deficit (–€21.6bn in 2025) now accumulates at the very front of the curve with no amortization vehicle left (the 2020 organic law capped CADES; the 2026 €15bn transfer barely dents it). France's Tier 1 WAM is shortening structurally — compounding exactly the pathology the main report flags (44% bill issuance, wide refi gap).
- **Reconciliation:** 94% (AFT central securities) + ~12.5pp agencies ≈ 107%, vs Maastricht ~113%; the residual is local governments (~€250bn, almost all SFIL/AFL/bank loans — which is why French regions appear on no issuer list).
- **Spreads are quasi-sovereign and move with OATs, not against them** (20–30bp over OAT; explicit guarantee on Unédic COVID paper; CADES an EPA of the state). A flat spread add to the French curve is exactly right — no credit lever needed. CADES/social-deficit figures are hard published numbers; ACOSS year-ends are softest (±€10bn, CP program with large intra-year swings).

Italy — CDP postal + CDP bonds, FS, regions (Lazio, Lombardy, Piedmont, Sicily)

€324bn (2016) → €365bn (2025), CAGR ~1.3% — essentially flat; 88% of it is one item, CDP postal savings (~€320bn, ~15% of GDP). Weighted WAM ~2.4y, weighted spread ~12bp over BTPs.

- **The postal stock is putable at par, daily.** Buoni carry 4–20y contractual terms but redeem anytime at face value: effective WAM ~2y and state-dependent — when BTP yields spike, savers redeem buoni at par and buy BTPs (net outflows –€11bn in 2022 exactly as BTPs sold off). The funding drains precisely in the bear scenario: it looks termed-out and behaves like overnight money in stress. Extends the report's 'latent-acute' Italy argument to its quasi-sovereign funding.
- **A securities screen finds ~€45bn, not €365bn.** Buoni/libretti are retail savings certificates with no conventional ISIN market: only CDP EMTN (~€28bn) + FS (~€12bn) + regional sinkers (~€5bn) appear. Hardcode the postal stock from CDP's accounts for the contingent exposure.
- **Double-counting runs both directions.** CDP lends a large chunk of postal money back to the Treasury and local governments (already in the BTP stock), and ~€40–50bn of legacy pre-2001 buoni sit on the MEF's books inside official state debt. The truly additive contingent liability is ~€200–250bn; treat the full postal stock as a guarantee-exposure scenario rather than a base-case interest add — CDP services it from asset income, not the budget.
- **Regions are a non-event:** <€6bn of amortizing legacy sinkers, no issuance since the 2011-era derivative scandals; their real debt (Lazio ~€20bn) is MEF/CDP loans, netting against the point above. **Cost is cheap by design:** administered rates below the BTP curve (the China model in milder form) plus ~55bp distribution fees to Poste (€1.8bn/yr); all-in ≈ BTP-flat at short tenors.

General caveats: figures target ~90% accuracy per the project brief; hard published anchors are flagged in each CSV's notes column, interpolations ±5–10%. Spreads are assumed flat adds at the stock's WAM point. 2026 rows are budget plans/projections. Sources: provincial public accounts and financing authorities (OFA, Finances Québec, Alberta TB, BC

MoF), state CFAs (TCorp, TCV, QTC, WATC) and RBA, China MoF and chengtou research (CCX, S&P;, Caixin), Destatis, Sénat/CADES/Unédic, CDP/Poste disclosures, NBC/RBC/TD/Scotiabank fiscal research.